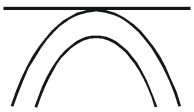
SOLUTION

SECTION–I

1. Ans. (A)



$f'(x) \le 0$

$\Rightarrow (a+2)x^2 - 2ax + 3a \le 0 \ \forall \ x \in R$

$\Rightarrow a + 2 < 0 \ \& \ D \le 0$

2. Ans. (B)

$f'(x) = m - e^{-x} \le 0 \ \forall \ x \in R$

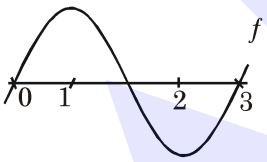
$\Rightarrow \quad m \le 0$

3. Ans. (B)

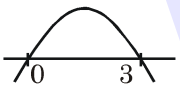
$$\left|y\frac{dy}{dx}\right| = k \ \Rightarrow \ ydy = \pm kdx$$

$$\Rightarrow \quad \frac{y^2}{2} = \pm kx + C$$

4. Ans. (A)



Min. no.of roots of $f(x) = 0$ is 3.



Min. no.of roots of $g(x) = 0$ is 2.

given equation is $f''g + f'g' = 0$

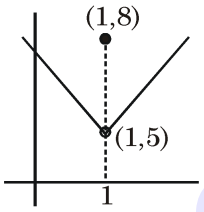
i.e. $(f'(x).g(x))' = 0$

$\because$ $f'(x) = 0$ has minimum 2 roots

$\therefore$ $f'(x).g(x) = 0$ has minimum 4 roots. (so using Rolles theorem)

$\Rightarrow (f'(x).g(x))' = 0$ has minimum 3 roots.

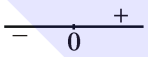
5. Ans. (A)



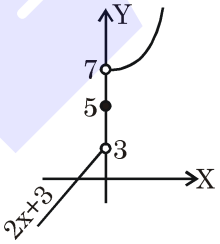
Clearly x = 1 is a point of local maxima.

6. Ans. (C)

$$f'(x) = \frac{x\left(x^2+4\right)}{\left(1+x^2\right)^{3/2}},$$

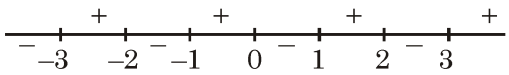


7. Ans. (B)



8. Ans. (B)

$f' = 2x(x^2-1)(x^2-4)(x^2-9)$



$f(x)$ have 7 critical points and $f'' = 0$ have 3 positive and 3 negative solutions

f' is odd function $\Rightarrow f'''$ is also an odd function so $f'''(0) = 0$

as there are 5 roots of $f'''(x) = 0 \Rightarrow$ it will have 2 positive and 2 negative roots.

9. Ans. (D)

$\because$ $f'(x) = -3x^2 - 2$

$\therefore$ $f(x)$ is decreasing

$\Rightarrow f(f(x)) < f(-x)$

$\Rightarrow f(x) > -x \Rightarrow 30 - 2x - x^3 > -x$

$\Rightarrow x^3 + x - 30 < 0$

$\Rightarrow (x-3)(x^2+3x+10) < 0 \Rightarrow x < 3$

$\therefore$ Non negative x = 0,1,2

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10. Ans. (C)

Let length = breadth = x and height = y

Given, $x^2 + 4xy = 48$ ...(i)

and volume $V = x^2y = x^2\left(\frac{48 - x^2}{4x}\right)$

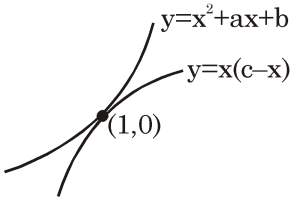
$\Rightarrow V = \frac{1}{4}(48x - x^3) \Rightarrow \frac{dv}{dx} = \frac{1}{4}(48 - 3x^2) = 0$

$\Rightarrow x = \pm 4$

Now (i) $\Rightarrow y = \frac{48 - 16}{16} = 2$

$\therefore V_{max} = 4^2.2 = 32m^3$.

11. Ans. (A,B,C,D)



$0 = 1 + a + b$ .....(i)

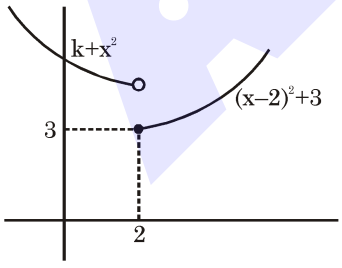
$0 = c - 1$ .....(ii)

& $\left(\frac{dy}{dx}\right)_{(1,0)} = \left(\frac{dy}{dx}\right)_{(1,0)}$

$\Rightarrow \quad 2 + a = c - 2$ .....(iii)

$a = -3, b = 2, c = 1$

12. Ans. (A,B,C)



13. Ans. (A,D)

(A) $f(x) > 0, f'(x) > 0$ & $f''(x) > 0$

(B) $f(x) > 0, f'(x) < 0$ & $f''(x) > 0$

(C) $f(x) < 0, f'(x) < 0$ & $f''(x) < 0$

(D) $f(x) < 0, f'(x) > 0$ & $f''(x) < 0$

14. Ans. (A,D)

From given condition

$f(x) = e^x$

Now $g(x) = \frac{3x}{e^x}$

$g'(x) = \frac{3(1 - x)}{e^x}$

(A) $g'\left(\ell n\sqrt{e}\right) = g'\left(\frac{1}{2}\right) = \frac{3}{2}.e^{-\frac{1}{2}}$

(B) $g'(x) = 0 \Rightarrow x = 1$

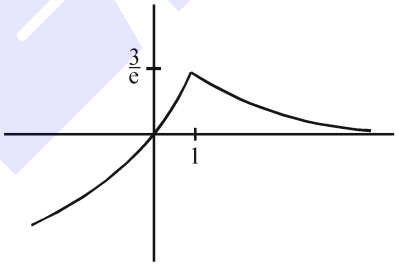
and $g'(1^-) = +ve$

$g'(1^+) = -ve$

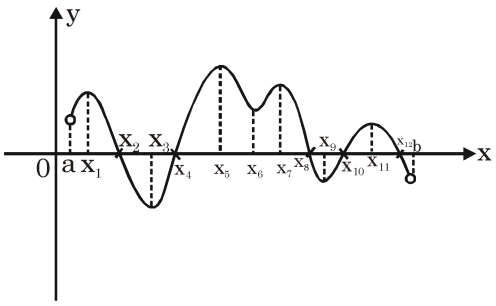
$\Rightarrow$ g(x) has maxima at x = 1

(C) x = 0 is not a critical point

(D) Graph of $g(x) = \frac{3x}{e^x}$



15. Ans. (D)



$f'(x)$ +ve to –ve at x_2, x_8, x_{12}

$\Rightarrow$ point of maxima

$f'(x)$ –ve to +ve at x_4, x_{10}

$\Rightarrow$ point of minima

$f''(x) = 0$ at $x_1, x_3, x_5, x_6, x_7, x_9, x_{11}$

$\Rightarrow$ Point of inflection

$f'(x)$ = zero at

$x_2, x_4, x_8, x_{10}, x_{12} \Rightarrow$ Stationary points

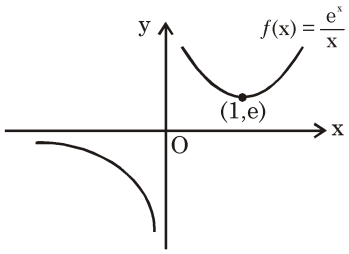
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SECTION–II

1. Ans. 0.50

a = 6, b = 2, c = 1, d = 2

2. Ans. 3.00



$$f'(x)=\frac{(x-1)e^x}{(x-0)^2}$$

$\therefore$ f is increasing in $(1, \infty)$

f is decreasing in $(-\infty, 1) - \{0\}$ & $x = 0$ is asymptote of curve

$\therefore$ $f_{min} = e$

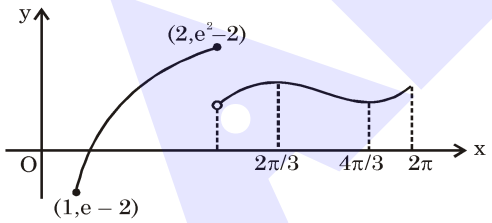
$\therefore$ Answer = 3

3. Ans. 0.73

$$f'(x)=\begin{cases} e^x-2, & 1<x<2 \\ 2\cos x+1, & 2<x<2\pi \end{cases}$$

In [1, 2] $f'(x) > 0$

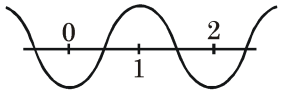
in $(2, 2\pi)$ $f'(x) = 0$ gives $\underset{\text{local max}}{\frac{2\pi}{3}}$, $\underset{\text{local min}}{\frac{4\pi}{3}}$



$\therefore$ least value occur at x = 1

$\therefore$ $\lambda = e - 2$

4. Ans. 0.25



$f(x) = 2x^4 - 8x^3 + 8x^2 - 1$

$f'(x) = 8x^3 - 24x^2 + 16x$

$= 8x(x^2 - 3x + 2)$

$= 8x(x - 1)(x - 2)$

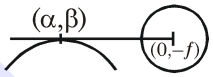
$f(0) = -1, f(1) = 1, f(2) = -1$

so $f(x)$ has 4 distinct roots.

5. Ans. 0.50

$xy' + y = 2x$

$$\Rightarrow \left.\frac{dy}{dx}\right|_{(\alpha,\beta)} = \frac{\beta+f}{\alpha-0}$$



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